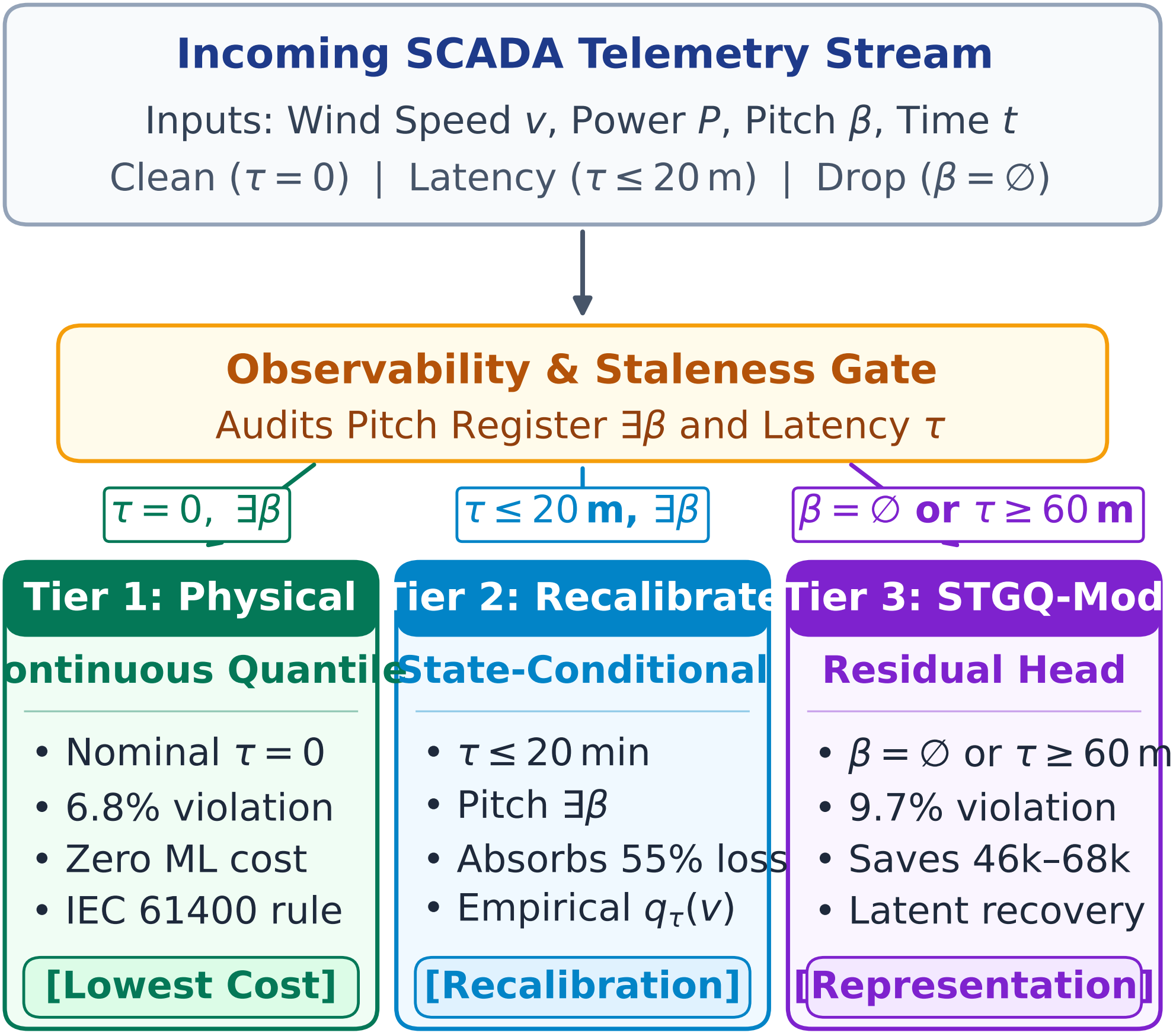
Incoming SCADA Telemetry Stream
Inputs: Wind Speed v, Power P, Pitch β, Time t
Clean ($\tau = 0$) | Latency ($\tau \leq 20$ m) | Drop ($\beta = \varnothing$)
Observability & Staleness Gate
Audits Pitch Register $\exists\beta$ and Latency τ
$\tau = 0,\ \exists\beta$
$\tau \leq 20$ m, $\exists\beta$
$\beta = \varnothing$ or $\tau \geq 60$ m
Tier 1: Physical
Continuous Quantile
• Nominal $\tau = 0$
• 6.8% violation
• Zero ML cost
• IEC 61400 rule
[Lowest Cost]
Tier 2: Recalibrate
State-Conditional
• $\tau \leq 20$ min
• Pitch $\exists\beta$
• Absorbs 55% loss
• Empirical $q_\tau(v)$
[Recalibration]
Tier 3: STGQ-Mod
Residual Head
• $\beta = \varnothing$ or $\tau \geq 60$ m
• 9.7% violation
• Saves 46k–68k
• Latent recovery
[Representation]